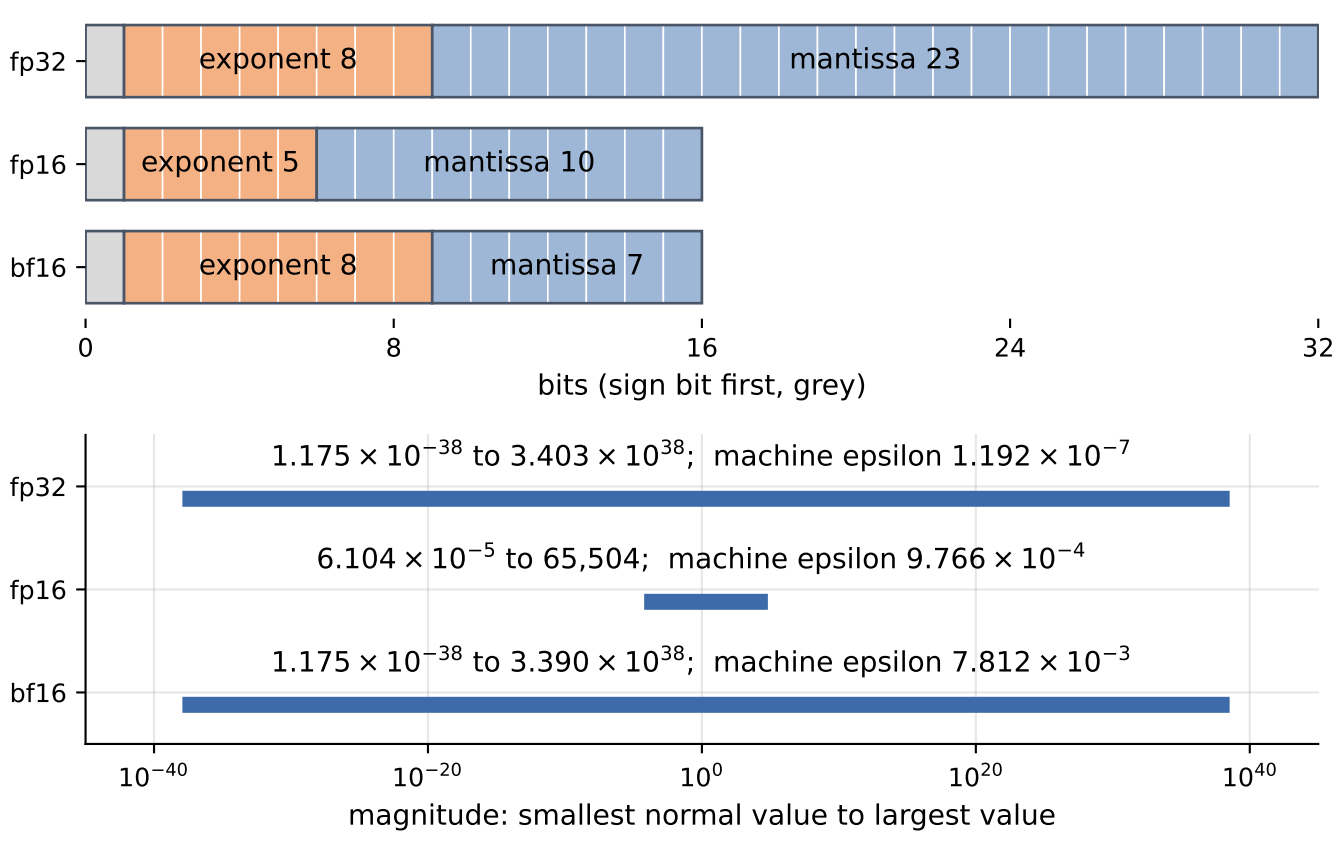
fp32
exponent 8
mantissa 23
fp16
exponent 5
mantissa 10
bf16
exponent 8
mantissa 7
0
8
16
24
32
bits (sign bit first, grey)
1.175×10^{-38} to 3.403×10^{38}; machine epsilon 1.192×10^{-7}
fp32
6.104×10^{-5} to 65,504; machine epsilon 9.766×10^{-4}
fp16
1.175×10^{-38} to 3.390×10^{38}; machine epsilon 7.812×10^{-3}
bf16
10^{-40}
10^{-20}
10^{0}
10^{20}
10^{40}
magnitude: smallest normal value to largest value